

Graphics Processing Unit (GPU)

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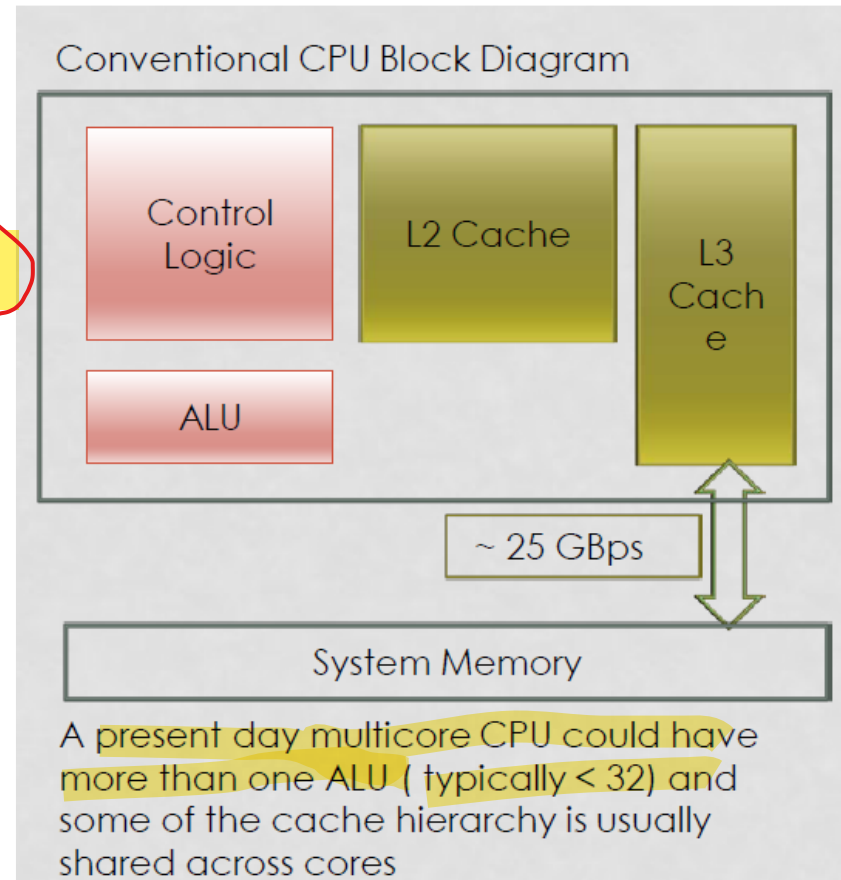
Course Title: Microprocessors and Assembly Language
Department of Computer Science and Engineering (CSE),
Islamic University of Technology (IUT), Gazipur.

Graphics Processing Unit (GPU)

- ▶ A specialized circuit designed to rapidly manipulate and alter memory
- ▶ Accelerate the building of images in a frame buffer intended for output to a display

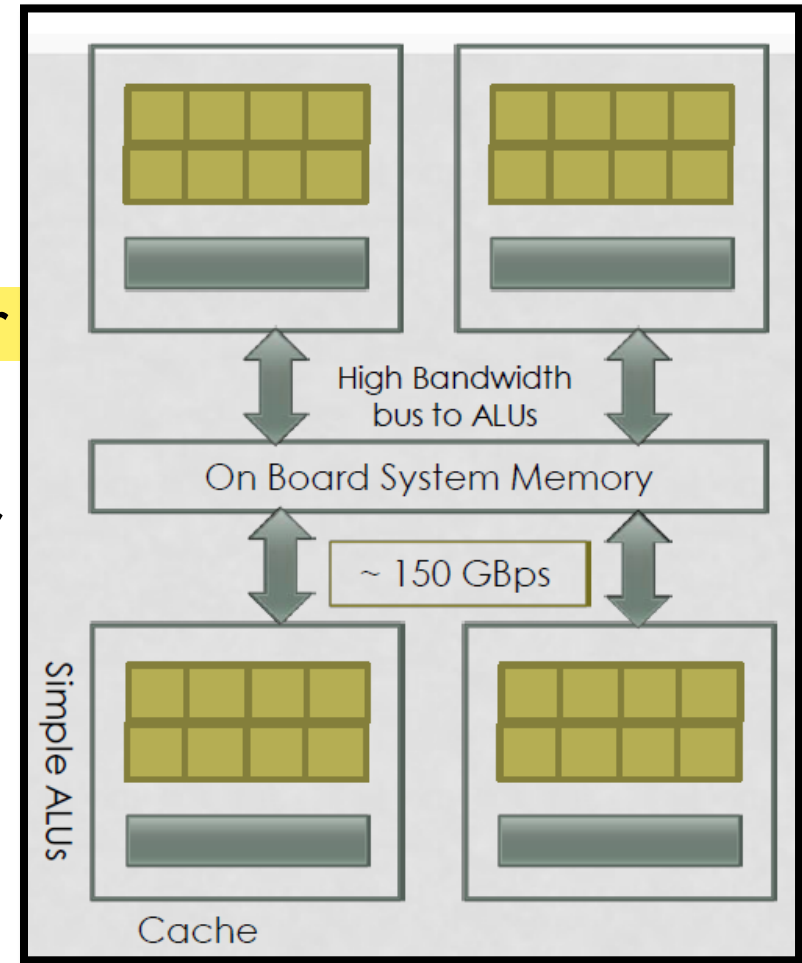
Conventional CPU Architecture

- ▶ Space devoted to control logic instead of ALU
- ▶ CPUs are optimized to minimize the latency of a single thread
 - ▶ Can efficiently handle control flow intensive workloads
- ▶ Multi level caches used to hide latency
- ▶ Limited number of registers are used due to smaller number of active threads



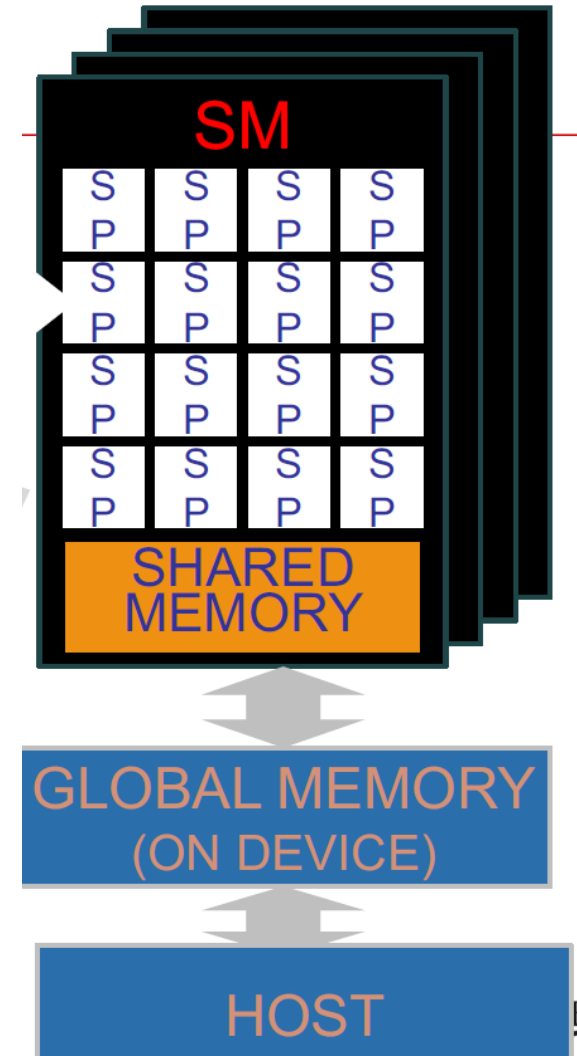
Modern GPU Architecture

- ▶ **Generic many core GPU**
 - ▶ Less space devoted to control logic and caches
 - ▶ Large register files to support multiple thread contexts
- ▶ **Low latency hardware managed for thread switching**
- ▶ Large number of ALU per “core” with small user managed cache per core
- ▶ Memory bus optimized for bandwidth
 - ▶ **~150 GBps bandwidth** allows us to service a large number of ALUs simultaneously



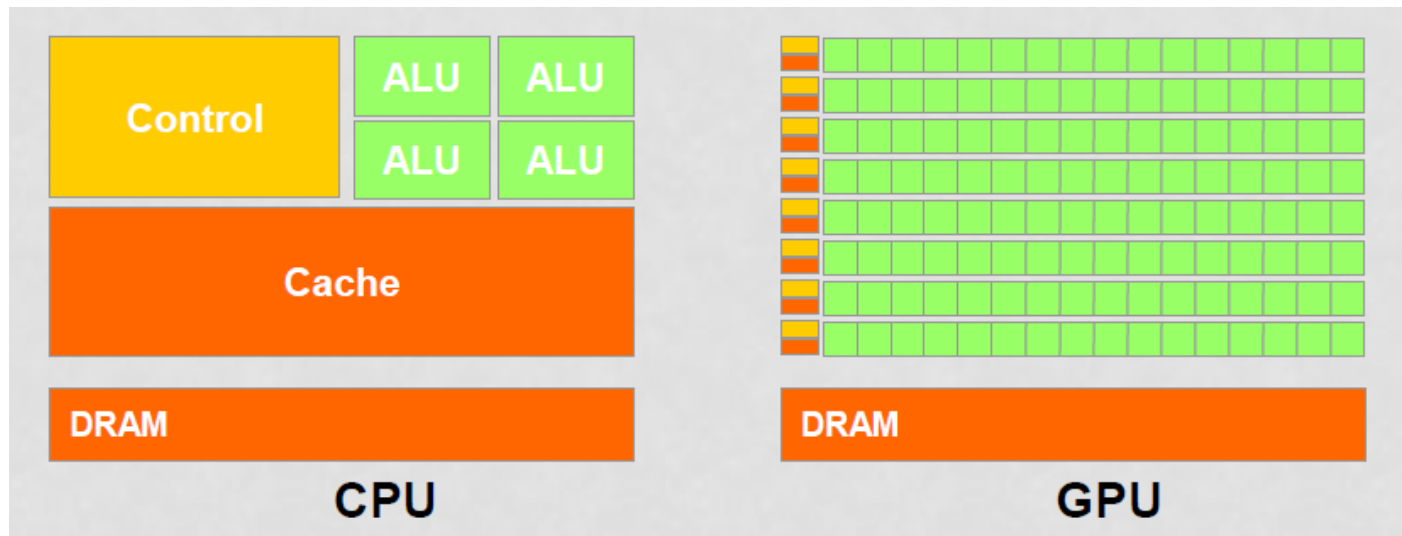
Modern GPU Architecture

- ▶ **SP:** Scalar Processors in 'CUDA core' architecture, each executes one thread
- ▶ **SM: Streaming multiprocessor**
 - ▶ 32xSP (or 16, 48 or more)
 - ▶ Fast local 'shared memory' (shared between SPs) 16 KB (or 64 KB)
- ▶ **Parallelization**
 - ▶ Decomposition to threads
- ▶ **Memory**
 - ▶ Shared memory, global memory
- ▶ **Enormous processing power**
- ▶ **Thread communication**
 - ▶ Synchronization, no interdependencies



Factors Behind GPU Revolutions

- ▶ The GPU is specialized for computing-intensive with highly parallel computation (exactly what graphics rendering is about)
- ▶ So, more transistors can be devoted to data processing rather than data caching and flow control



- ▶ The fast-growing video game industry exerts strong economic pressure that forces constant innovation

GPU Vendors



NVIDIA GPU Architecture

- ▶ **Compute Unified Device Architecture (CUDA) Framework**
 - ▶ A general purpose parallel computing architecture
 - ▶ A new parallel programming model and instruction set architecture
 - ▶ Leverages the parallel compute engine in NVIDIA GPUs
 - ▶ Software environment that allows developers to use C as a high-level programming language
- ▶ **CUDA Capable Devices: Tesla, Quadro, NVS, GeForce**

NVIDIA GPU Architecture

- ▶ Device = GPU = Set of multiprocessor
- ▶ Warp = A scheduling unit of up to 32 threads
- ▶ Multiprocessor = Set of processors & shared memory
- ▶ Kernel = GPU program (Typically OS)
- ▶ Grid = array of thread blocks that execute a kernel
- ▶ Thread block = group of SIMD threads that execute a kernel and can communicate via shared memory

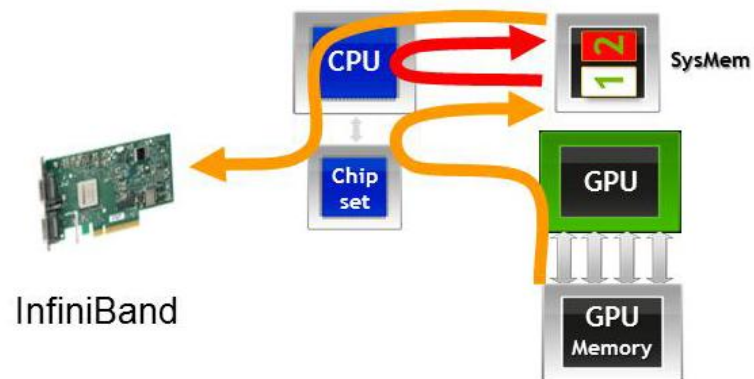
NVIDIA Architecture

► GPUDirect NVIDIA GPUDirect™ Accelerated Communication with Network And Storage Devices

Without GPUDirect

Same data copied three times:

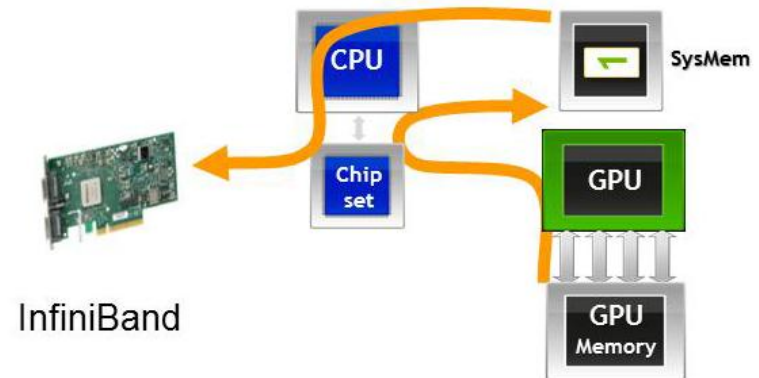
1. GPU writes to pinned systemmem1
2. CPU copies from system1 to system2
3. InfiniBand driver copies from system2



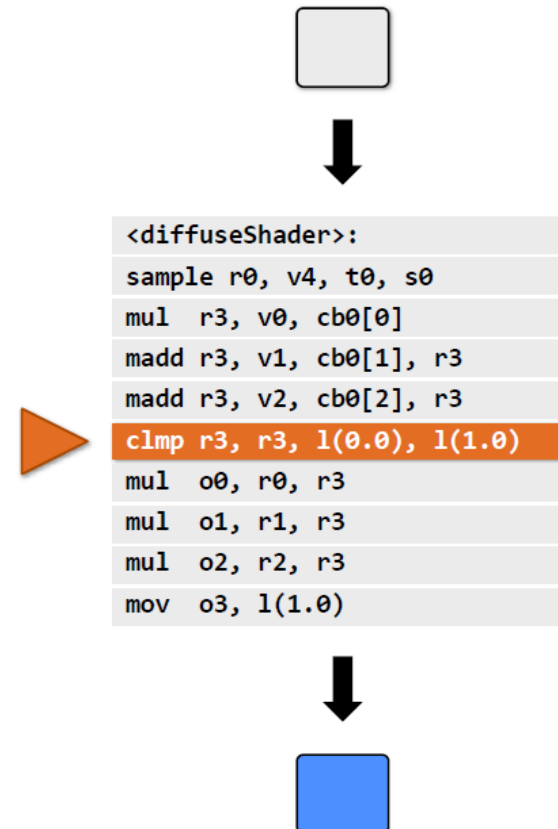
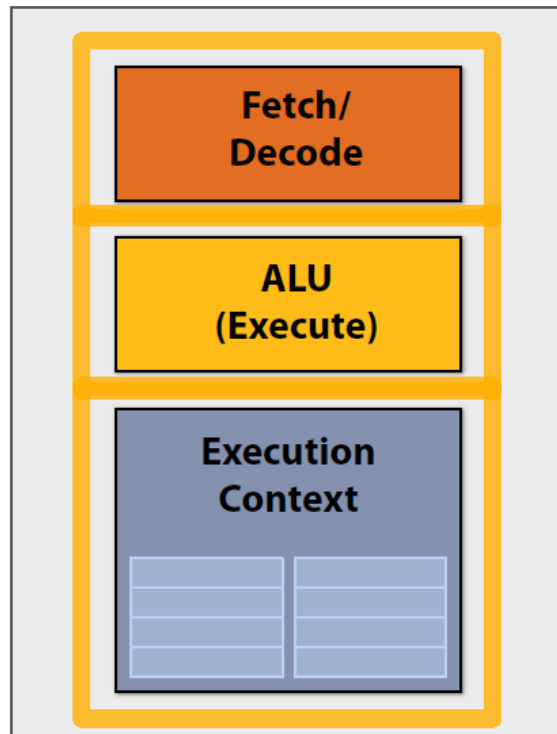
With GPUDirect

Data only copied twice

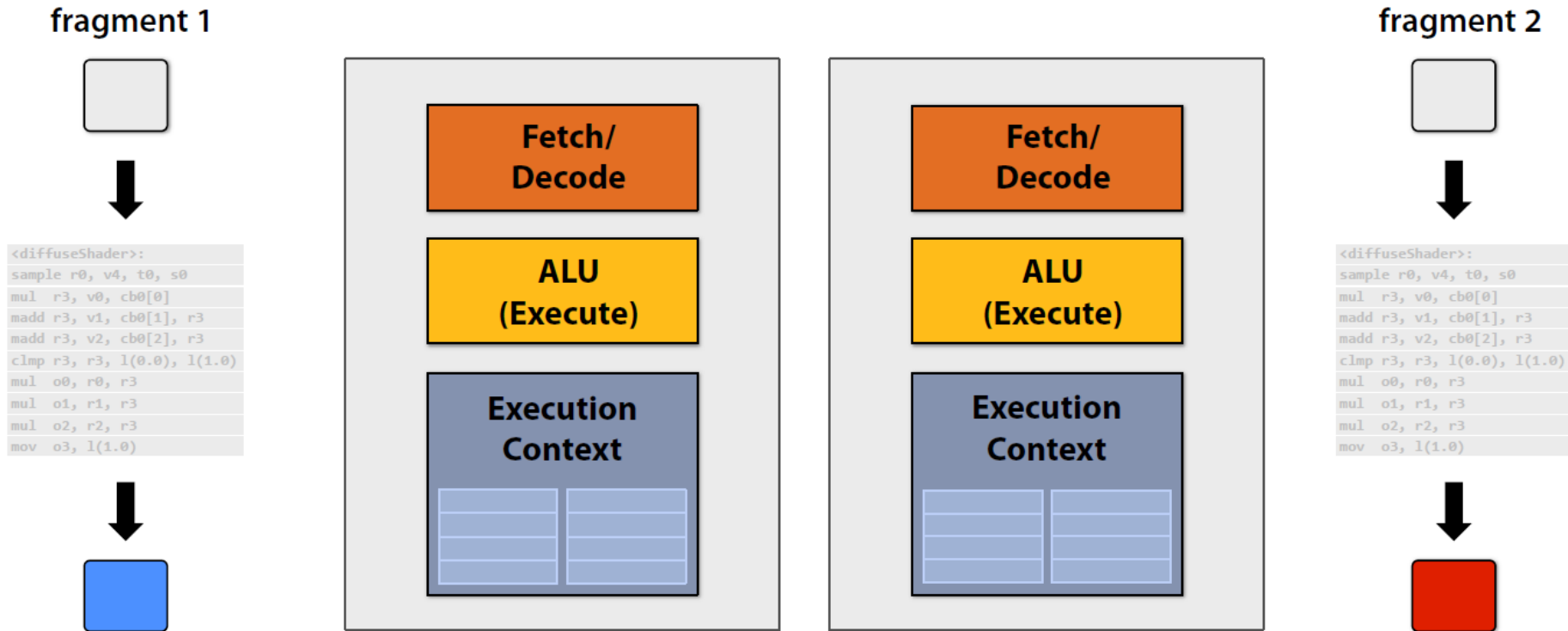
Sharing pinned system memory makes system-to-system copy unnecessary



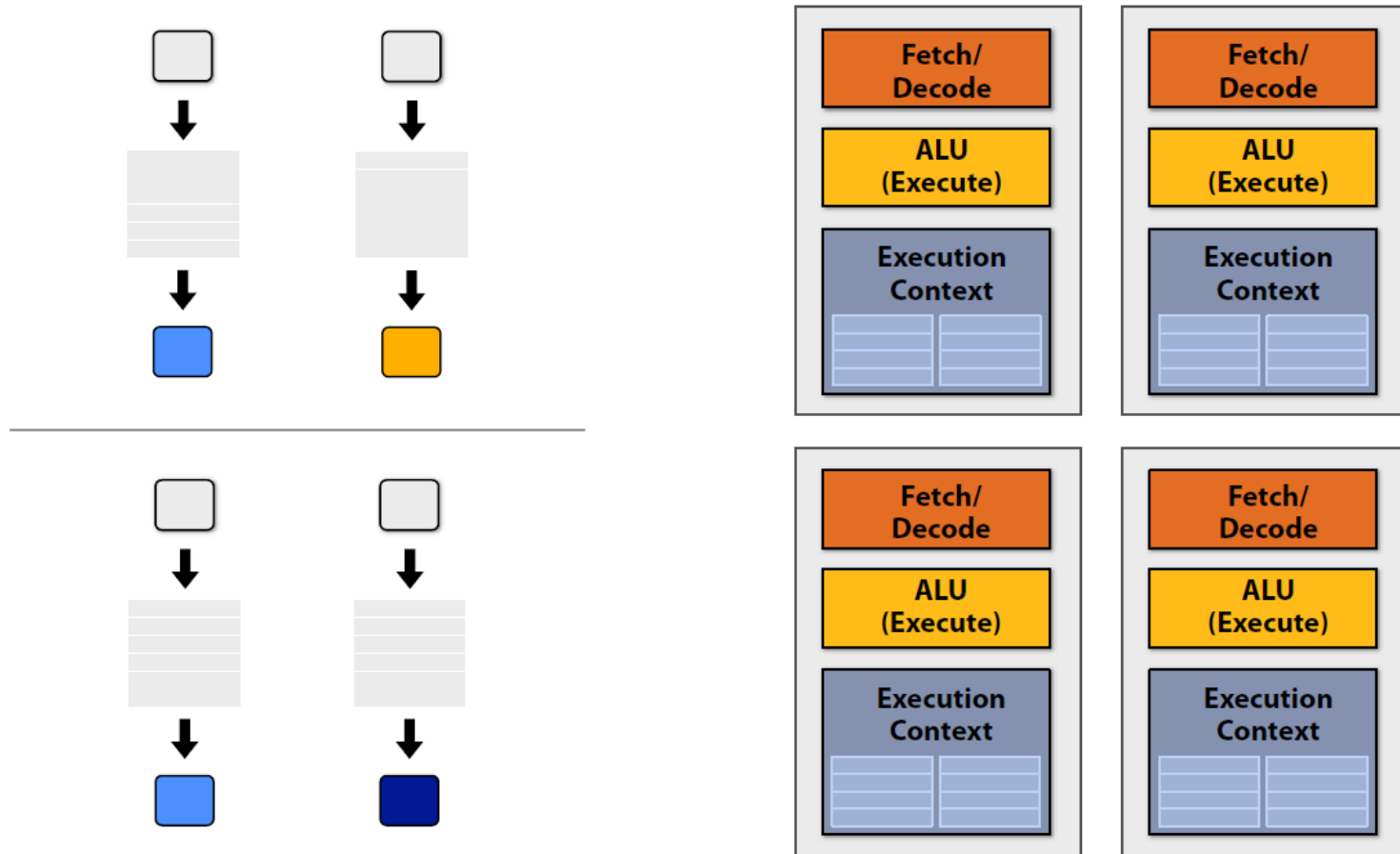
CPU



GPU (Two Cores)



GPU (Four Cores)



GPU vs. CPU



Intel Core i7 Quad Core

~100 GFLOPS peak
730 million transistors



AMD Radeon HD 5870

~2.7 TFLOPS peak
2.2 billion transistors

GPU vs. CPU

- ▶ Main advantage of CPU is caches
 - ▶ Fast single thread performance, but also helps with multithreaded apps
 - ▶ Disadvantage is complexity, limiting number of cores per chip
 - ▶ Also have fast synchronization
- ▶ Main advantage of GPU is high throughput
 - ▶ Each instruction for an SM executes on 8 scalar units (32 data elements)
 - ▶ Disadvantage is need to move data explicitly into (small) SM memory from large shared memory
 - ▶ Also have support for gather/scatter from memory and special functional units (e.g., texture sampling, math ops)
- ▶ Performance measurements for GPU assume data already in GPU memory
- ▶ Overall performance of GPU (geometric mean) is 2.5X of CPU

Thank You !!

